

1. $a = (4, 2, 0)$; $b = (1, 2, 2)$

$$\text{proj}_b a = \frac{a \cdot b}{b \cdot b} b = \frac{(4 + 4 + 0)}{(1 + 4 + 4)} (1, 2, 2)$$

$$= \left(\frac{8}{9}, \frac{16}{9}, \frac{16}{9} \right)$$

perpendicular component, $a_{\perp} = a - \text{proj}_b a$

$$= (4, 2, 0) - \left(\frac{8}{9}, \frac{16}{9}, \frac{16}{9} \right)$$

$$= \left(\frac{28}{9}, \frac{2}{9}, -\frac{16}{9} \right)$$

2. $v_1 = (1, 0, 1)$; $v_2 = (0, 1, 1)$; $x = (2, 3, 4)$

Gram-Schmidt, $u_1 = v_1 = (1, 0, 1)$

norm, $\|u_1\| = \sqrt{2}$

$$q_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{2}} (1, 0, 1)$$

$$u_2 = v_2 - \text{proj}_{u_1}(v_2) = (0, 1, 1) - \frac{1}{2} (1, 0, 1) = \left(-\frac{1}{2}, 1, \frac{1}{2}\right)$$

$$q_2 = \frac{u_2}{\|u_2\|} = \frac{1}{\sqrt{6}} (-1, 2, 1)$$

~~∴ Projection matrix is~~ $Q = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{pmatrix}$

Projection matrix, $P = Q Q^T = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{2}{3} \end{pmatrix}$

$$Px = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{2}{3} \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$$

$$= \left(\frac{5}{3}, \frac{8}{3}, \frac{13}{3} \right)$$